

REGULATION

PROJECT:

Regulatory texts are lengthy, highly technical, and require **strict factual accuracy**. General-purpose LLMs often **hallucinate** and fail to capture legal nuance, making them unreliable for regulatory reasoning.

Goal: accurate, grounded Q&A over banking regulations.

Solution: Domain-aware RAG + FT models.

The Data:

- Official Bank of Israel banking & financial regulations.
- Synthetic Q&A generation to ensure coverage and consistency.
- Questions designed to test precision, grounding, and edge cases.



SYNTHETIC DATA GENERATION

THE PROMPT

```
prompt_template = """
You are a strict data generator for banking regulation compliance.
Based **STRICTLY** on the provided context, generate **4 distinct** training examples.

Context:
"{context}"

Task:
Generate a JSON LIST containing 2 'Yes' and 2 'No' Question-Answer pairs.

Requirements:
1. **question**: Practical question by a bank employee (e.g., "Am I allowed to...", "Must we report...").
2. **answer**: "Yes" or "No".
3. **citation**: **CRITICAL**: This must be a **VERBATIM COPY-PASTE** from the text.
   - Extract the sentence EXACTLY as it appears in the text, including punctuation and weird spacing.
   - **DO NOT** rephrase, summarize, or fix grammar.
   - **DO NOT** combine two different sentences into one.
   - If the sentence is cut off in the context, copy only what is visible.
4. **explanation**: A short, professional rationale (1 sentence).
   - **STYLE**: Write as a factual statement or principle.
   - **FORBIDDEN**: DO NOT refer to "the text", "the passage", "this section", "the document", or "it says".
   - **BAD EXAMPLE**: "The text mentions that risk assessments are vital."
   - **GOOD EXAMPLE**: "Risk assessments are vital to ensure long-term stability."
5. **source_details**: Format: "Header, Section X" (or just Header).

Output format: Return ONLY a valid JSON LIST of objects.
"""
```

SAMPLE EXAMPLE

```
"question": "Are we permitted to apply credit policies only at the level of individual banking corporations?",
"context": "Supervisor of Banks: Proper Conduct of Banking Business [10] (11/23) \nCredit Risk Management \nPage 311-1",
"answer": "No",
"citation": "They shall be applied on a banking group basis, subject to the provisions of law, and at the level of indi",
"explanation": "Credit policies must be applied on both a banking group basis and at the individual corporation level."
"source": "311-17, Supervisor of Banks, Page 311-17"
```

INTERACTIVE INTERFACE

? Question (or 'exit'): Is it acceptable to implement credit policies without a written policy statement?

Verdict: No

Quote: The credit policy statement shall include, inter alia:

Source: 311_et.pdf, Page 11

Explain: A written policy statement is essential for effective credit risk management.

? Question (or 'exit'): Must we ensure a linkage between the depositing of a check drawn on a foreign bank and a transaction with a banking corporation in

Verdict: Yes

Quote: before the clearing transaction takes place, the banking corporation shall ensure the existence of a linkage between the depositing of the check and

Source: 411_et.pdf, Page 32

Explain: A linkage between check deposits and transactions is required for compliance.

? Question (or 'exit'): Can I transfer 1 million\$ to my lecturer?

Verdict: N.A

Quote: N.A

Source: N.A

Explain: The regulation regarding the transfer of funds does not address the specific scenario of transferring money to a lecturer.

? Question (or 'exit'): exit

MODELS & METRICS



Models Evaluated:

- **Llama + MiniLM** – strong general baseline.
- **Llama + LegalBERT** – domain-aware embeddings.
- **SaulLM + MiniLM** – legal LLM with generic retrieval.
- **SaulLM + LegalBERT** – full legal-domain stack.



Metrics Used:

- **Verdict Accuracy** – correct regulatory conclusion.
- **Hit@K** – relevant source retrieved in top-K.
- **Hallucination Rate** – percentage of definitive answers provided when the context contains no relevant information.

| NOVELTY & ACHIEVEMENTS

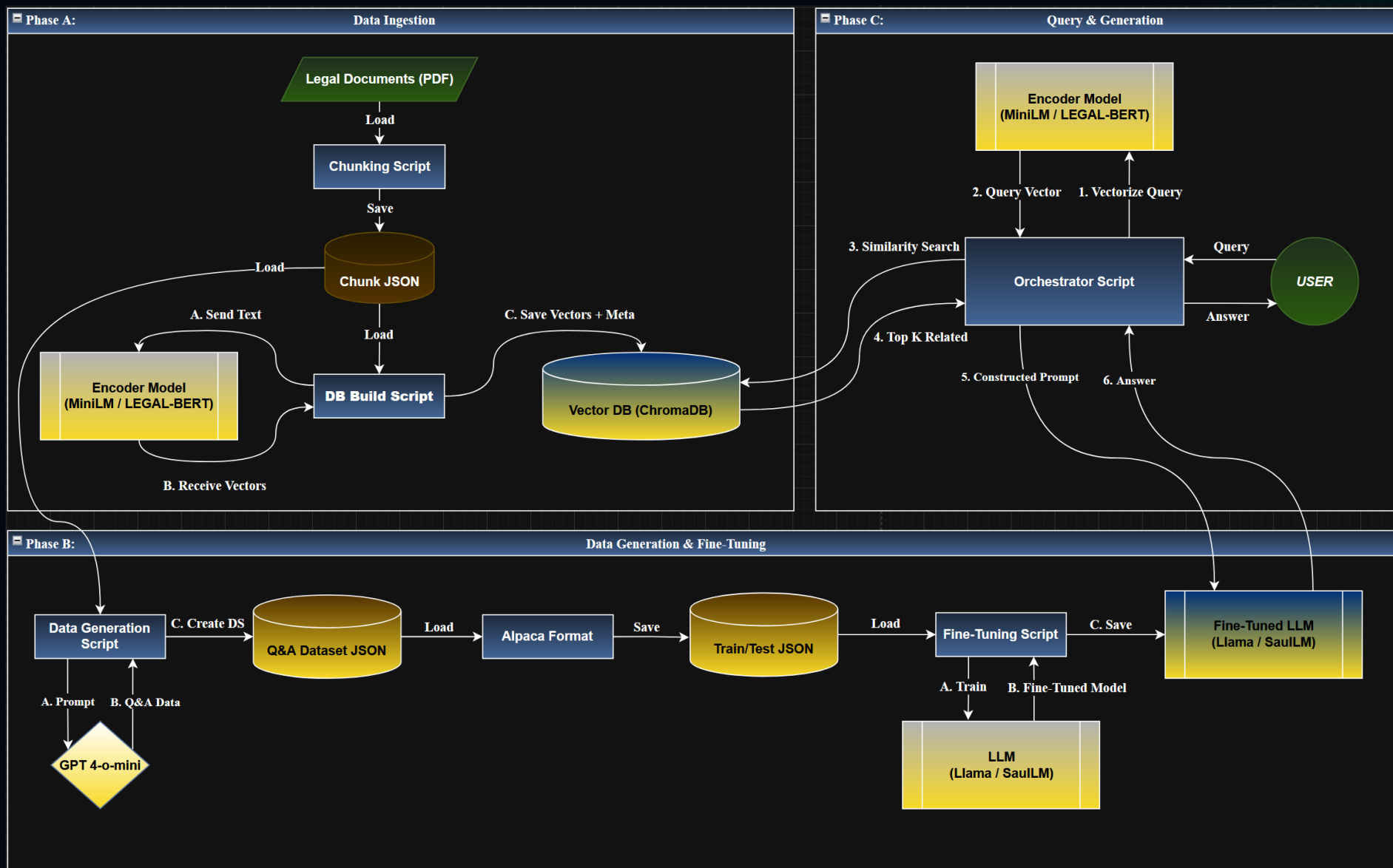
Project Novelty:

- Systematic comparison of legal vs general model stacks in a regulatory RAG setting.
- Introduced citation-level evaluation for regulatory correctness.
- Combined synthetic legal Q&A generation with real regulatory texts.
- Focused on trustworthiness, not just language fluency.

Project Achievements:

- Built a full end-to-end RAG system for regulatory question answering.
- Benchmarked general vs legal LLMs and embeddings under identical conditions.
- Designed a quantitative evaluation framework beyond generic accuracy.
- Improved answer grounding and source alignment in regulatory Q&A.

PIPELINE



METHODOLOGY

Part 1

Fine Tuning:

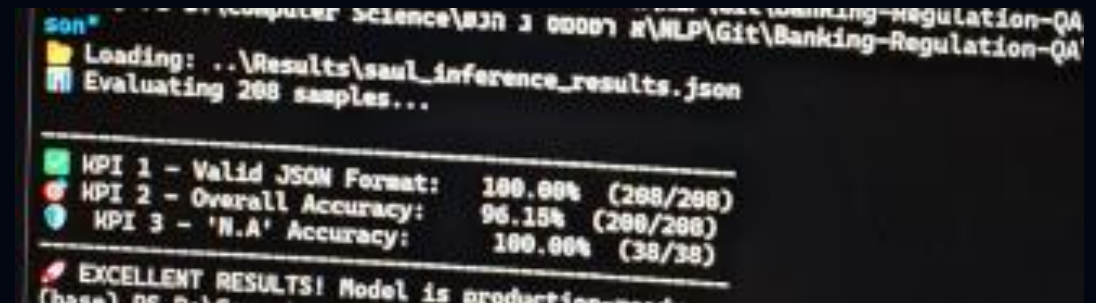
Updating the model itself to optimize it for the task.

- **Domain Specialization:-** Enhanced the model's regulatory expertise
- **Robustness to Negatives** - Trained the model on hard and soft negative samples to improve its ability to identify irrelevant information and avoid false positives.
- **Structured Output Optimization:** Fine-tuned the model to ensure consistent and valid JSON formatting

- **Model Selection:** Evaluated and compared **Llama 3.1** and **Saul-7B** (pre-trained on the legal domain).

- **Dataset Construction:** Compiled a dataset of **2,000+ examples**, including **15-20% negative samples** to enhance robustness and prevent hallucinations.

- **FT KPI'S:**



```
son* ... Computer Science\msh 1 00007 n\MLP\Git\Banking-Regulation-QA
Loading: ..\Results\saul_inference_results.json
Evaluating 208 samples...

KPI 1 - Valid JSON Format: 100.00% (208/208)
KPI 2 - Overall Accuracy: 96.15% (200/208)
KPI 3 - 'N.A' Accuracy: 100.00% (38/38)

EXCELLENT RESULTS! Model is production-ready.
```


| FINE-TUNNING INPUT & OUTPUT

ALPACA FORMAT INPUT (TRAIN)

```
"instruction": "Below is an instruction that describes a task, paired with an input that provides further context. Write a res  
"output": "{ \"answer\": \"Yes\", \"citation\": \"payment for said transaction is to be charged by means of a payment card belo
```

OUTPUT EXAMPLE (INFERENCE)

```
"question": "You are a banking regulation expert. Use the provided context to answer the question in JSON format. If the answer is  
"ground_truth": "{\n  \"answer\": \"Yes\", \n  \"citation\": \"An appropriate limit system should flag credit that has been app  
"model_prediction": "{\n  \"answer\": \"Yes\", \n  \"citation\": \"An appropriate limit system should flag credit that has been
```


METHODOLOGY

Part 2

RAG:

Providing the model with external data to ensure factual accuracy.

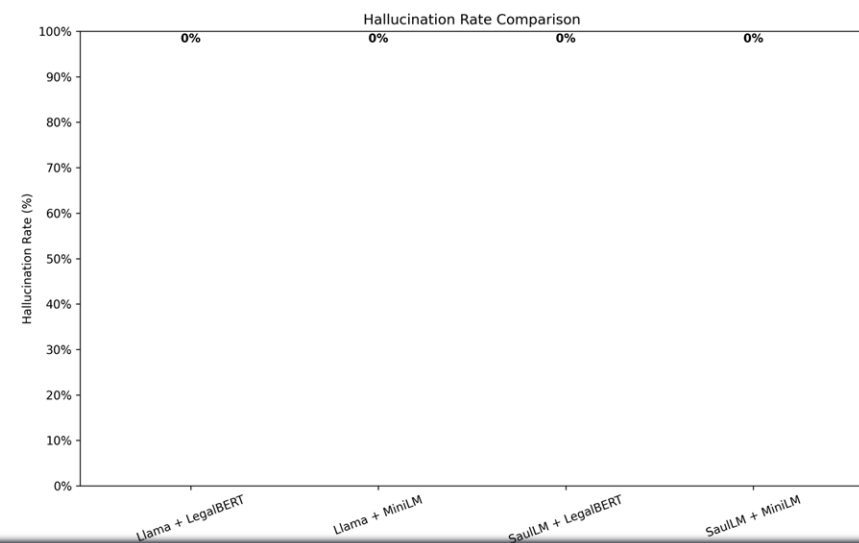
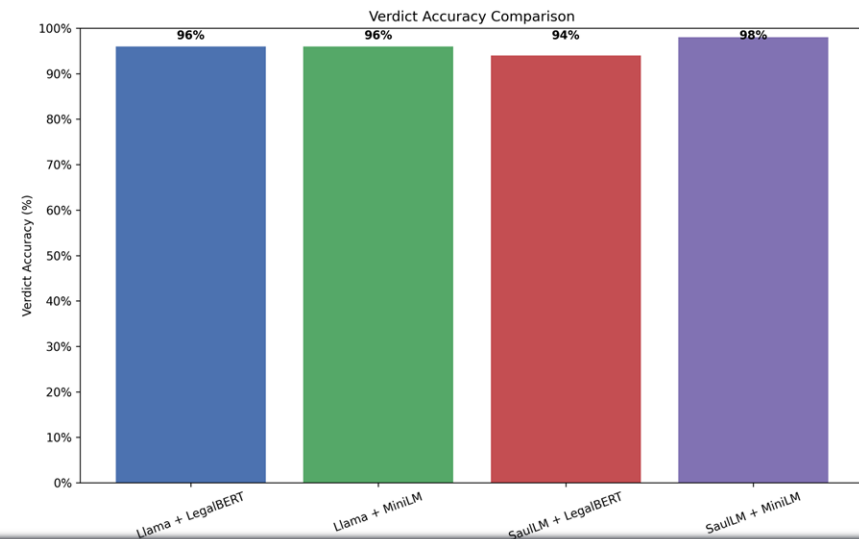
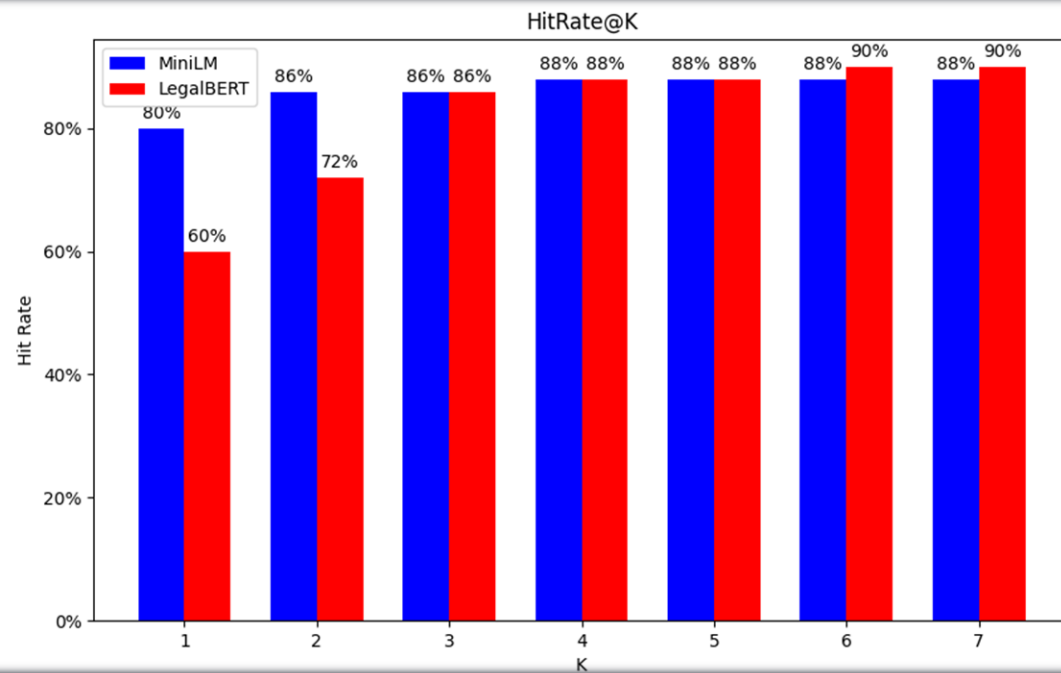
- **Context Injection:** Supplies relevant regulatory text for every query.
- **Evidence-Based:** Grounds all answers in **official sources** (B.O.I) to eliminate hallucinations.
- **Dynamic Knowledge:** Accesses the latest documents without the need for retraining.

- **Vector Database:** Engineered a vector database using **ChromaDB**, consisting of **518 text chunks** extracted from official regulations.
- **Embedding Model:** Compared **all-MiniLM** and **LegalBERT** to determine the most effective embedding model for banking and legal terminology.
- **Retrieval Evaluation:** Measured **Hit@K (HitRate)**

```
Starting Retrieval Benchmark for: [MINILM]
Connecting to DB at: /content/drive/MyDrive/Regulation/Data/RAG_db_all
Checking 50 questions...
100% 50/50 [00:18<00:00, 2.71it/s]

=====
🎯 HIT RATE RESULT: MINILM
=====
Valid Questions:      50
Successful Hits:      43
Hit Rate @3:          86.00%
=====
📄 Full report saved to: /content/drive/MyDrive/Regulation/Results/retrieval
```

RESULTS



|CONCLUSION

Looks Good... But Can Be Better:

- Minor hallucinations are still present.
- The model identifies the correct chunk in most cases, but consistency is not guaranteed.
- In a high-stakes domain like this, even small errors are unacceptable.

Things We (Painfully) Learned:

- Each model comes with its own pros and cons.
- Hallucination handling needs to be significantly stricter.
- A diverse set of KPIs is needed to properly evaluate model quality.

Where Do We Go From Here?

- Experiment with additional models and architectures.
 - Introduce more rigorous validation and consistency checks in the RAG pipeline.
 - Add a faithfulness metric using an LLM judge to verify citation-explanation alignment.
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